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Photovoltaic expansion occupied large staple cropland with minimal yield reduction in China

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During the United Nations Climate Summit in September 2025, China announced plans to expand wind and solar power capacity to more than six times the 2020 level by 2035 [1]. As a key technology, solar photovoltaics (PV) requires intensive land use [2], which creates pronounced tensions with cropland [3]. In response, Chinese authorities have remained cautious about PV development on cropland [4]. Although previous studies have examined the potential risks of unauthorized land occupation and potential agricultural yield losses [5], the spatial overlap between PV deployment and cropland, as well as the resulting impacts on crop yields, remains unclear.

Here, we developed an integrated approach to map PV installations on cropland across the Chinese mainland from 2010 to 2022 and to quantify their impacts on crop yields. The methodology comprised three main steps (Fig. 1a). First, croplands equipped with PV systems and their corresponding staple crop types were identified (Texts S1, S2 online). Second, whether cultivation continued after PV installation was determined. For sites where cultivation had ceased, a 100% yield loss was assigned. For sites where cultivation continued, the potential yield loss was quantified using the ground coverage ratio (GCR), defined as the ratio of PV panel area to total system area [6]. The relationship between GCR and crop yields, established through a meta-analysis of field experiments [6], was used as the key metric. Finally, the GCR of PV power stations in China was inverted from satellite imagery, and this relationship was applied to assess large-scale impacts on crop yields (Text S2.2 online).

The key findings are summarized as follows:

1. *Rapid expansion of PV installations on cropland in China.* Our analysis reveals that more than 25% of the total installed PV area in China, approximately around 400 thousand hectares (ha), was located on cropland. In particular, the PV area on cropland

increased 1667-fold, from 60 ha in 2010 to 100 thousand ha in 2022 (Fig. S1b online). This rapid expansion was begun in 2013, with the annual increase rising steadily from 22.5 thousand ha in 2013 to a peak of 46 thousand ha in 2017. The policy was introduced in 2013, which emphasized nationwide PV development (Fig. S2 online), likely contributing to this growth. In 2017, China introduced a policy prohibiting PV development on permanent basic cropland. As a result, the annual increase in PV area on cropland in 2018 declined by about one-third relative to 2017. Despite these regulatory constraints, the total PV area on cropland continued to grow substantially, with annual increases ranging from 20 thousand ha to 32 thousand ha between 2018 and 2022 (upper left panel of Fig. S1b online). Meanwhile, PV expansion on cropland exhibited pronounced regional heterogeneity. Overall, PV installations on cropland were concentrated in central and eastern China (Fig. S1a online). In 2010, 60% of the PV installations on cropland were located on the Huang-Huai-Hai Plain and showed a scattered distribution pattern (Fig. S1c online). After 2017, more than 80% of PV installations on cropland became concentrated in three major agricultural zones, namely the Huang-Huai-Hai Plain, the Middle-lower Yangtze Plain, and the Loess Plateau, owing to the rapid nationwide deployment of PV, particularly in central and eastern China [5].

2. *Spatial heterogeneity of PV on staple-crop cropland.* Our approach identified two distinct land-use patterns for PV installations on cropland: continued cultivation or complete cropland abandonment (Text S2.3 online). Overall, abandoned land accounted for 52% of PV-occupied cropland, with corresponding proportions of 62%, 59%, and 37% for rice, maize, and wheat, respectively (Fig. S3 online). On cropland where cultivation continued, accounting for 48% of the total, crop yield losses were closely associated with PV GCR. Rice showed an average GCR of 42%, slightly higher than maize at 40% and wheat at 39%. This difference was accompanied by substantial spatial heterogeneity in PV GCR across the major agricultural zones of China (Fig. 1b, Fig. S4 online). Higher PV GCR values, ranging from 48.6% to 66%, were mainly observed in the Middle-lower Yangtze Plain and the Sichuan Basin,

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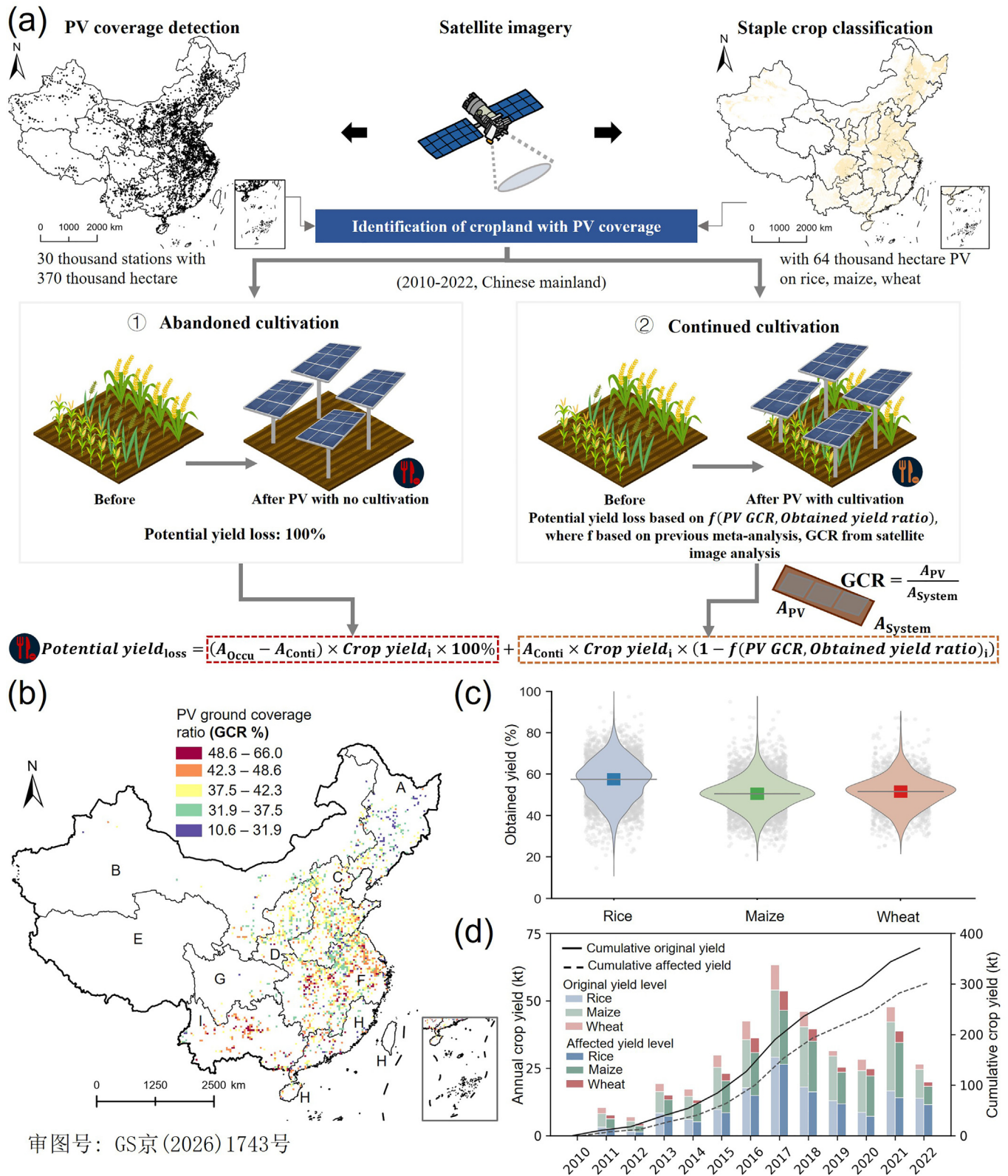


Fig. 1. Assessment of potential crop yield impacts from PV deployment on cropland in China. (a) Conceptual framework for assessing potential crop yield loss from PV expansion on cropland. (b) Spatial distribution of PV ground coverage ratios (GCR) and variations across nine agricultural zones for the three staple crops, A-I, A: Northeast China Plain, B: Northern arid and semiarid region, C: Huang-Huai-Hai Plain, D: Loess Plateau, E: Qinghai-Xizang Plateau, F: Middle-lower Yangtze Plain, G: Sichuan Basin and surrounding regions, H: Southern China, I: Yunnan-Guizhou Plateau. (c) Obtained yield percentages (relative to pre-PV baseline yields) of three staple crops under varied PV GCR, with the line width in the graph indicating the density of data points at each obtained yield percentage value. (d) Annual and cumulative yield levels of three staple crops before and after PV deployment.

where rice and maize cultivation predominates. In contrast, the North China Plain showed lower PV GCR values, ranging from 10.6% to 37.5%. These results indicate marked variation in GCR among geographical regions and crop types, which could lead to diverse impacts on crop yields.

3. *Limited crop yield losses associated with PV installations.* Our results indicate that, although the area of PV deployment on cropland increased substantially, the estimated crop yield loss of 300 kt accounted for only 0.05% of the total annual production of the three staple crops in China (Fig. 1d), which averages more than 600 million tons (Mt) [7,8]. This limited impact could be related to the variation in obtainable yield percentages, quantified relative to yields before PV installation, which ranged from 20% to 80% across the three staple crops (gray points in Fig. 1c). Among these crops, rice showed better yield retention, maintaining 60% of baseline productivity, whereas maize and wheat each maintained approximately 50%. The estimated peak annual yield losses occurred in 2017, reaching 26 kt for rice, 20 kt for maize, and 7.2 kt for wheat (Fig. 1d).

In summary, this study presents a systematic approach for assessing the effects of PV installations on crop yields in China. Our analysis shows that, by 2022, one-quarter of PV deployment was located on cropland. Despite the expansion of PV installations on cropland, the estimated impacts on crop yields remained limited. As a global leader in photovoltaic development, China is making significant strides toward global decarbonization goals, underpinned by the accelerated rollout of utility-scale solar installations [9], under which half of renewable energy is expected to come from solar power. This ambition is likely to intensify tensions over cropland use [5,10]. Under these circumstances, synergies between solar energy production and cropland conservation need to be better utilized in order to meet energy demand while protecting agricultural land [11]. Some case studies have shown that PV shading can alleviate drought stress in crops and even increase yields [12], highlighting both the potential for PV deployment on cropland in arid regions [13] and the importance of investigating regional differences in environmental and yield outcomes.

Based on our findings, some tailored policies rather than a one-size-fits-all approach are recommended for China to achieve the synergistic expansion of renewable energy and food production. The methodology presented here enables rapid, large-scale assessment of PV impacts on crop yields at the national scale. It therefore provides a technical foundation for the refined management of cropland PV installations on cropland and offers a practical pathway to maximize food-energy co-benefits from croplands in the near future.

Conflict of interest

The authors declare that they have no competing interests.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scib.2026.07.044>.

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